PBL Synopsis
PCS253

Project Title: Institutional Energy Usage Analysis and Prediction System

Problem Id: 23

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Problem Description:

Energy management in large institutions is a significant challenge due to fluctuating occupancy and varying environmental conditions. This project addresses the need for accurate energy consumption forecasting using Machine Learning. In real-life scenarios, institutions face high operational costs and carbon footprints; by predicting energy demand based on factors like time of day, student count, and ambient temperature, administrators can optimize HVAC systems, schedule appliance usage efficiently, and implement cost-saving measures. This regression-based system aims to transform raw institutional data into actionable insights, ensuring sustainability and financial efficiency.

Steps of Implementation:

1. **Data Collection:** Gathering numerical datasets containing institutional records such as timestamps, occupancy levels (number of students), weather data (temperature/humidity), and power consumption metrics.
2. **Data Preprocessing:** Cleaning the dataset by handling missing values and outliers. This

phase includes normalization/scaling of numerical features and encoding any categorical variables to ensure consistency for the regression model.

3. **Exploratory Data Analysis:** Using statistical summaries and visualizations to identify patterns (e.g., peak hours), correlations between temperature and energy, and the impact of student density on usage.
4. **Feature Selection & Engineering:** Applying techniques such as Principal Component Analysis (PCA) or Forward/Backward Selection to identify the most influential features, reducing dimensionality and improving model accuracy.
5. **Model Selection & Training:** Choosing appropriate regression algorithms (e.g., Linear Regression, Random Forest Regressor, or XGBoost) and training them on the processed data.
6. **Evaluation & Validation:** Testing the model on unseen data using metrics like Mean Squared Error (MSE) and R-squared to ensure the model generalizes well.
7. **Final Model:** Finalizing the predictive system for deployment or reporting.

Expected Modules / Libraries to be used:

- **Pandas:** Used for data manipulation, cleaning, and analysis using DataFrame structures.
- **NumPy:** Essential for performing high-level mathematical functions on multi-dimensional arrays and matrices.
- **Matplotlib / Seaborn:** Used for creating static, animated, and interactive visualizations like heatmaps and line plots.
- **Scikit-Learn:** The primary library for implementing machine learning algorithms, feature selection, and evaluation metrics.
- **SciPy:** Used for scientific computing and technical computing, specifically for statistical analysis during EDA.

Expected Evaluation Metrics and Plots to be used:

- **Mean Absolute Error (MAE) & Root Mean Squared Error (RMSE):** Used to quantify the average magnitude of prediction errors in energy units (kWh).
- **R-squared (R^2) Score:** To determine how well the independent variables explain the variance in energy consumption.
- **Correlation Heatmap:** To visualize the relationship strength between temperature, occupancy, and energy use.
- **Actual vs. Predicted Plot:** A scatter plot to visually inspect how closely the model's predictions align with real-world data points.
- **Residual Plot:** To check for non-linearity and ensure the errors are randomly distributed.

Required Topics from the Subject:

- **Linear Regression & Multi-variate Regression:** Understanding the mathematical foundation for predicting continuous numerical values based on multiple inputs.
- **Data Normalization & Standardization:** Learning how to rescale data to prevent features with larger magnitudes from dominating the model.
- **Feature Selection Methods:** Understanding how to use statistical tests to remove redundant data and prevent overfitting.
- **File Handling in Python:** Knowledge of reading and writing CSV or JSON files to import datasets and export model results.

Platform Required:

- Jupyter Notebook / Google Colab (for interactive EDA and model development).
- Python 3.x Environment.

Books and Link Sources:

1. **Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow** by Aurélien Géron.
2. **Introduction to Machine Learning with Python** by Andreas C. Müller & Sarah Guido.
3. **Scikit-Learn Documentation:** https://scikit-learn.org/stable/
4. **Kaggle Datasets (Energy Category):** https://www.kaggle.com/datasets